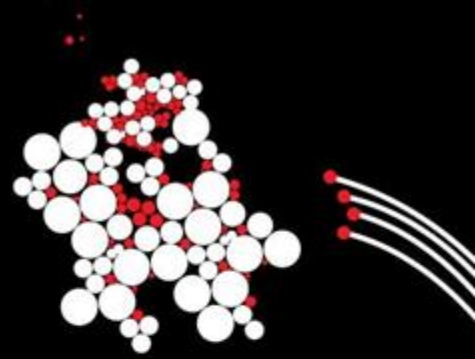
UNIVERSITY OF TWENTE.



P4 Lab

Advanced Networking

Friday Sep 5, 2025



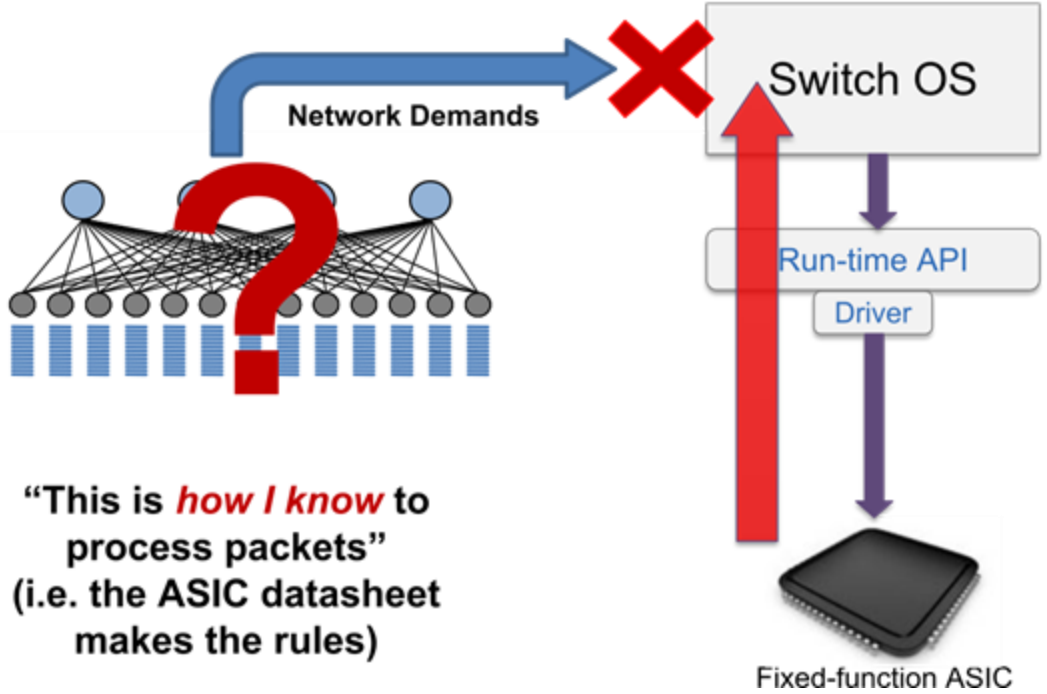
Shyam Krishna Khadka



Overview

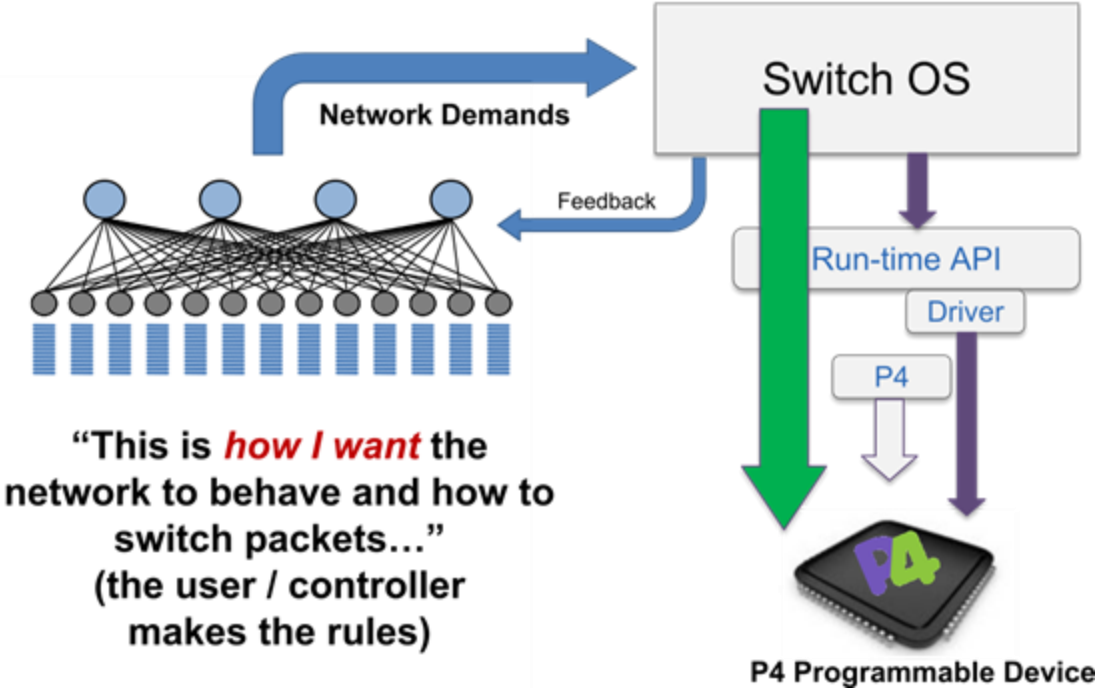
- Introduction
- Architectural Overview
- Programming in P4
- Lab Assignments

Introduction
Bottom-up



"This is *how I know* to process packets" (i.e. the ASIC datasheet makes the rules)

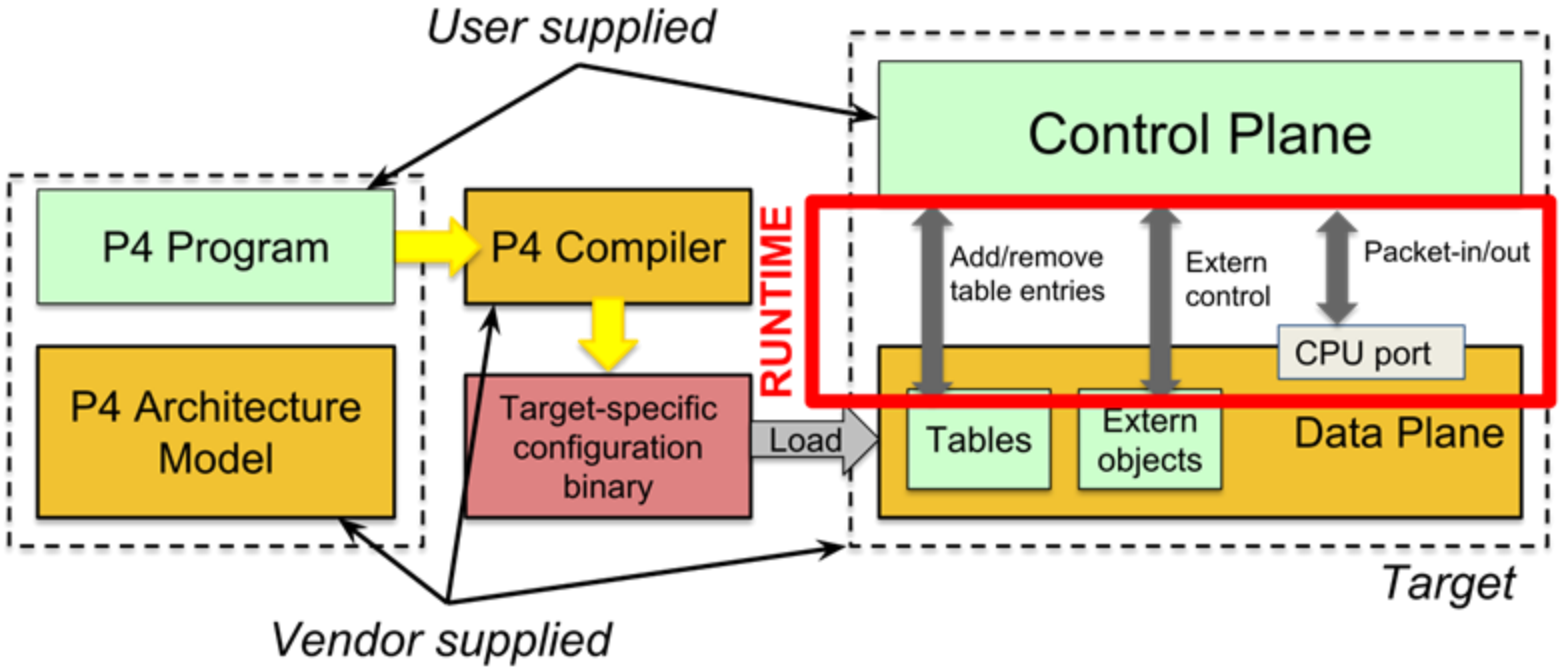
Introduction
Top-down



p4.org

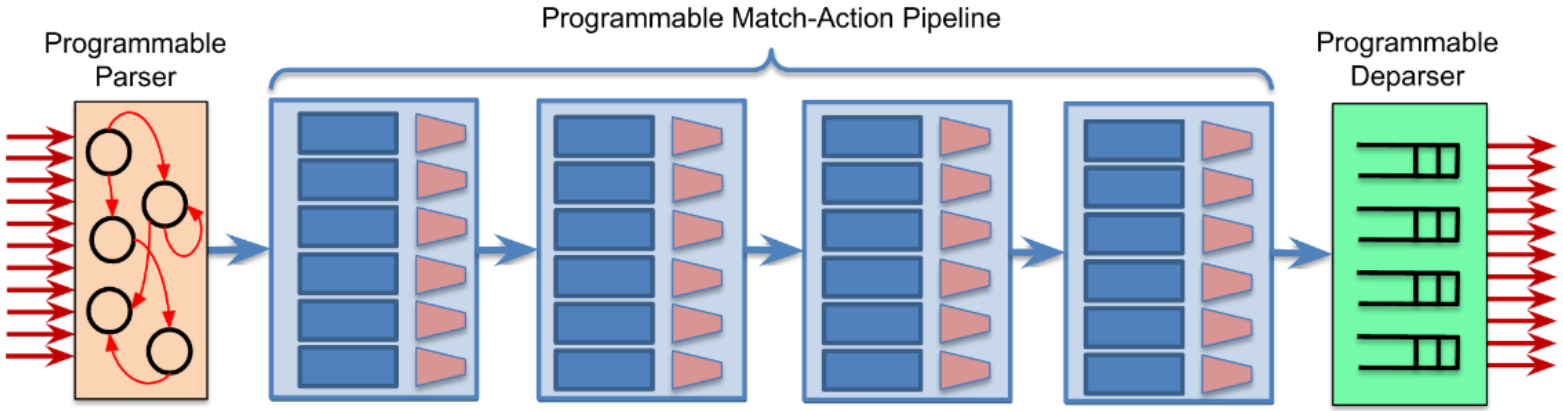
Architectural Overview

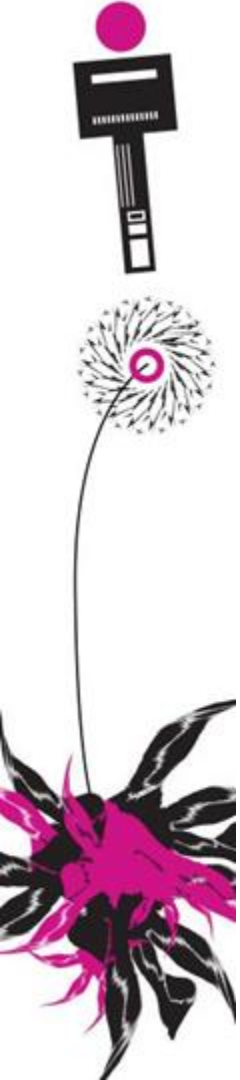
P4



Architectural Overview

PISA: Protocol Independent Switch Architecture





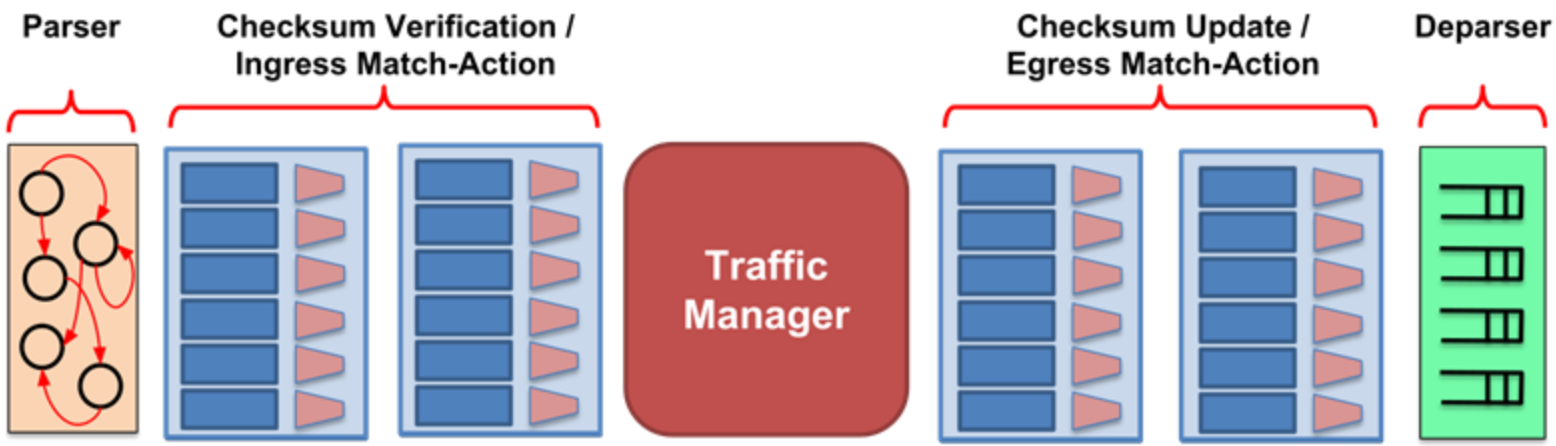
Architectural Overview

bmv2 Switch

- Software switch: P4 **Target**
- Used in the lab assignments
- For developing, testing and debugging
- V1Model: P4 **Architecture** model for bmv2

Architectural Overview

V1Model stages



Programming in P4

V1Model stages

```
#include <core.p4>
#include <v1model.p4>
/* HEADERS */
struct metadata { ... }
struct headers {
  ethernet_t   ethernet;
  ipv4_t       ipv4;
}
/* PARSER */
parser MyParser(packet_in packet,
                out headers hdr,
                inout metadata meta,
                inout standard_metadata_t smeta) {
  ...
}
/* CHECKSUM VERIFICATION */
control MyVerifyChecksum(in headers hdr,
                         inout metadata meta) {
   ...
}
/* INGRESS PROCESSING */
control MyIngress(inout headers hdr,
                  inout metadata meta,
                  inout standard_metadata_t std_meta) {
  ...
}
```

```
/* EGRESS PROCESSING */
control MyEgress(inout headers hdr,
                 inout metadata meta,
                 inout standard_metadata_t std_meta) {
  ...
}
/* CHECKSUM UPDATE */
control MyComputeChecksum(inout headers hdr,
                          inout metadata meta) {
  ...
}
/* DEPARSER */
control MyDeparser(inout headers hdr,
                   inout metadata meta) {
  ...
}
/* SWITCH */
V1Switch(
  MyParser(),
  MyVerifyChecksum(),
  MyIngress(),
  MyEgress(),
  MyComputeChecksum(),
  MyDeparser()
) main;
```

Programming in P4

Metadata and V1Model Standard Metadata

```
struct standard_metadata_t {
    bit<9>  ingress_port;
    bit<9>  egress_spec;
    bit<9>  egress_port;
    bit<32> clone_spec;
    bit<32> instance_type;
    bit<1>  drop;
    bit<16> recirculate_port;
    bit<32> packet_length;
    bit<32> enq_timestamp;
    bit<19> enq_qdepth;
    bit<32> deq_timedelta;
    bit<19> deq_qdepth;
    bit<48> ingress_global_timestamp;
    bit<32> lf_field_list;
    bit<16> mcast_grp;
    bit<1>  resubmit_flag;
    bit<16> egress_rid;
    bit<1>  checksum_error;
}
```

- **`ingress_port`** - the port on which the packet arrived
- **`egress_spec`** - the port to which the packet should be sent to
- **`egress_port`** - the port on which the packet is departing from (read only in egress pipeline)

Programming in P4
Parsing

- `extern`: interface for functionality provided by switch vendor
  - Similar to abstract classes/methods in OOP
- State machine
- Transitions
  - `select`: change state
  - `accept`: finish parsing

Parsing

```
// packet_in: extern for input packet
extern packet_in {
  void extract<T>(out T hdr);
  void extract<T>(out T hdr,in bit<32> n);
  T lookahead<T>();
  void advance(in bit<32> n);
  bit<32> length();
}

// parser: begins in special "start" state
state start {
  transition parse_ethernet;
}

// User-defined parser state
state parse_ethernet {
  packet.extract(hdr.ethernet);
  transition select(hdr.ethernet.type) {
    0x800: parse_ipv4;
    default: accept;
  }
}
```

Programming in P4

Actions, Control Flow & Tables

Actions

```
// Inputs provided by control-plane
action set_next_hop(bit<32> next_hop) {
  if (next_hop == 0) {
    metadata.next_hop = hdr.ipv4.dst;
  } else {
    metadata.next_hop = next_hop;
  }
}

// Inputs provided by data-plane
action swap_mac(inout bit<48> x,
                inout bit<48> y) {
  bit<48> tmp = x;
  x = y;
  y = tmp;
}

// Inputs provided by control/data-plane
action forward(in bit<9> p, bit<48> d) {
  standard_metadata.egress_spec = p;
  headers.ethernet.dstAddr = d;
}

// Remove header from packet
action decap_ip_ip() {
    hdr.ipv4 = hdr.inner_ipv4;
    hdr.inner_ipv4.setInvalid();
}
```

Tables

```
table ipv4_lpm {
  key = {
    hdr.ipv4.dstAddr : lpm;
    // standard match kinds:
    // exact, ternary, lpm
  }
  // actions that can be invoked
  actions = {
    ipv4_forward;
    drop;
    NoAction;
  }
  // table properties
  size = 1024;
  default_action = NoAction();
}
```

Control Flow

```
apply {
  // branch on header validity
  if (hdr.ipv4.isValid()) {
    ipv4_lpm.apply();
  }
  // branch on table hit result
  if (local_ip_table.apply().hit) {
    send_to_cpu();
  }
  // branch on table action invocation
  switch (table1.apply().action_run) {
    action1: { table2.apply(); }
    action2: { table3.apply(); }
  }
}
```

Programming in P4

Actions, Control Flow & Tables

```
#include <core.p4>
#include <v1model.p4>
struct metadata {}
struct headers {}

parser MyParser(packet_in packet, out headers hdr,
   inout metadata meta,
   inout standard_metadata_t standard_metadata) {
    state start { transition accept; }
}

control MyIngress(inout headers hdr, inout metadata meta,
   inout standard_metadata_t standard_metadata) {
    action set_egress_spec(bit<9> port) {
        standard_metadata.egress_spec = port;
    }
    table forward {
        key = { standard_metadata.ingress_port: exact; }
        actions = {
            set_egress_spec;
            NoAction;
        }
        size = 1024;
        default_action = NoAction();
    }
    apply {   forward.apply();   }
}
```

```
control MyEgress(inout headers hdr,
   inout metadata meta,
   inout standard_metadata_t standard_metadata) {
    apply {  }
}

control MyVerifyChecksum(inout headers hdr, inout metadata
meta) {   apply { }   }

control MyComputeChecksum(inout headers hdr, inout metadata
meta) {   apply { }   }

control MyDeparser(packet_out packet, in headers hdr) {
    apply { }
}

V1Switch( MyParser(), MyVerifyChecksum(), MyIngress(),
MyEgress(), MyComputeChecksum(), MyDeparser() ) main;
```

Key	Action ID	Action Data
1	set_egress_spec ID	2
2	set_egress_spec ID	1

Programming in P4
Deparsing

- Emit headers in front of payload
- Watch the order!

Deparsing

```
// packet_out: extern for output packet
extern packet_out {
  void emit<T>(in T hdr);
}

apply {
  // insert headers into pkt if valid
  packet.emit(hdr.ethernet);
}
```

Lab Assignments

Repository

- https://gitlab.utwente.nl/m7717102/p4-labs-2025
- Fork of P4 language tutorials

Lab Assignments

Virtual Machine

- All necessary tools installed
  - Atom (with *Markdown Preview* package)
  - Wireshark
- Repository cloned, but make sure to pull the latest version

Lab Assignments

Virtual Machine

- Alternatively:
  - You can work in the host machine by making the repo a *shared folder.* You still need to run your code in the VM.
  - https://docs.oracle.com/en/virtualization/virtualbox/6.0/user/sharedfolders.html
    - Make sure *Auto-mount* and *Make Permanent* are checked
    - Make sure *Read-only* is NOT checked
  - Recommended for VSCode users: *p4-lang* by Zhanghan Wang

Lab Assignments

What you need to do

- Assignment 1
  - Basic Forwarding
  - Basic Tunneling
- Assignment 2
  - P4Runtime
- Assignment 3
  - Firewall
- Assignment 4
  - Load balancing
  - Controlled load balancing

Lab Assignments

How to carry them out

- Follow tutorial instructions (in the README files)
- Add comments to P4 code briefly explaining
  - What the code does
  - Why you did it that way
  - Parts of the cheat sheet that you used
- Only use the cheat sheet, do not use the answers (obviously)
- Upload P4 code to Canvas

Lab Assignments

Signing off

- Demonstrate your code and its behavior
- Briefly explain what you did
- Might ask more in-depth questions

Lab Assignments

Sessions

- Monday 30th September: Sign-off session #1
- Monday 21th October: Sign-off session #2

Grading:

- Pass if everything signed off on Monday 21th October



Lab Assignments

Tips

- Assume you need around **16 hours** to do all the assignments

  ⇒ around half the work needs to be done outside the lab sessions
- Try to have the **Assignments 1 & 2** finished at the **first lab session**
- Fully read what you have to implement before actually writing any code
- Understand the files you have to edit
- In the VM, if *Backspace* suddenly does not work anymore:
  use *CTRL* + *Backspace*

Links

P4 resources: http://P4.org/learn

Assignment Repo: https://gitlab.utwente.nl/m7717102/p4-labs-2024

V1Model source code and docs:

https://github.com/p4lang/p4c/blob/main/p4include/v1model.p4